

MOHAMED FAWZI YOUSEF MOHAMED

AI Solutions Architect & Full-Stack Engineer

Cairo, Egypt | +249 91 559 2263 (WhatsApp/Roaming) |
dr.mohamed.fawzi.y@gmail.com

GitHub: <https://github.com/WD-FAWZI> | Portfolio: <https://mf-portfolio-v5-final.vercel.app>

PROFESSIONAL SUMMARY A Medical Doctor (MBBS) transitioned into a Software Engineer and AI Solutions Architect (9 months post-graduation; all projects self-initiated and independently deployed). A self-directed engineer with a demonstrated rapid acquisition of complex technical domains, leveraging a clinical diagnostic mindset to engineer secure, scalable systems. Specializes in modern frontend ecosystems and AI architecture (Local LLMs, Voice Agents).

TECHNICAL SKILLS

- **Identity & Security:** OAuth 2.0, OIDC, SAML 2.0 (Conceptual), JWT, Supabase RLS.
- **Frontend:** React, Next.js, TypeScript, Vite, Tailwind CSS, Three.js, Monorepo Architecture.
- **Backend & Infra:** Node.js, Python, FastAPI, Supabase, Firebase, Vercel, RESTful APIs.
- **AI & Architecture:** Local LLMs (Ollama), Voice AI Agents (Vapi, ElevenLabs), Prompt Caching, Multi-Agent Workflows, Computer Vision (CV) pipelines.
- **Languages:** Arabic (Native), English (Fluent), Japanese (Beginner – actively studying).

ENGINEERING PROJECTS & EXPERIENCE

Itminan Health (December 2025) | Product Designer & Full-Stack Developer

- Engineered a secure full-stack clinical platform using Supabase (PostgreSQL), strictly enforcing Row Level Security (RLS) and leveraging managed OAuth 2.0 / JWT session protocols.
- Developed an interactive clinical dashboard with click-responsive visualizations for real-time tracking of dynamic symptoms.
- Automated clinical workflows via structured PDF reports using TestSprite AI to maintain high code reliability and automated testing coverage.

H.I.M (Voice AI Conversational Agent) | AI Solutions Architect, February 2026

- Developed a low-latency, real-time voice AI agent tailored for Sudanese dialect recognition, achieving a response latency of <500ms.
- Orchestrated complex external API integrations (Vapi, ElevenLabs) to achieve human-like conversational fluidity.
- Resolved critical challenges in real-time audio processing, ensuring continuous state management and high session reliability.

AI-Driven EdTech Assessment Tool | AI & Full-Stack Developer | Completed March 2026

- Prototyped and deployed an educational assessment platform for generating accurate, context-aware medical questionnaires.
- Implemented Prompt Caching and dynamic model switching, reducing API latency by ~40% and optimizing token costs.
- Executed successful User Acceptance Testing (UAT), receiving high praise for precision and speed from initial users.

HACKATHONS & PROJECTS

Mistral AI Hackathon (Feb 28 - Mar 1, 2026) - "ER Zero Latency" | Solo AI Solutions Architect

- Engineered a zero-latency Multimodal AI co-pilot for Emergency Room triage using passive CCTV surveillance.
- Programmed a CV pipeline (MediaPipe, OpenCV) to detect clinical distress signals and correlate visual data with EHRs for ACS prediction.
- Designed an event-driven 'Low-Power Standby' logic (Eco-Mode), reducing GPU consumption by 85%.
- Achieved a sub-100ms response time from visual trigger to final diagnostic alert.

EDUCATION

Bachelor of Medicine, Bachelor of Surgery (MBBS) - May 2025 Ibn Sina University (ISU), Khartoum, Sudan

- Completed degree with a concurrent self-directed focus on software engineering and AI architecture.